

Summary

AI Robotics Engineer with an M.S. in Electrical Engineering, focused on ROS 2, Isaac Sim / Isaac Lab, and Isaac GROOT for robot control, VLA, and Digital Twin data workflows. Experienced in end-to-end integration across perception, visual servoing, force-guided contact search, data processing, policy inference, frontend/backend platforms, and simulation / real-world validation.

Key Results

3-4x imitation-learning / GROOT-VLA training-data throughput	70% → 90% cable-insertion success in 10 trials
---	---

Experience

- AMTRAN TECHNOLOGY CO., LTD. | AI Robotics EngineerMay 2025 - Present | New Taipei, Taiwan
- Built ROS 2 control and simulation workflows for industrial and collaborative robot platforms with NVIDIA Isaac Sim / Isaac Lab, MoveIt 2, and cuMotion for motion planning, collision handling, pick-and-place, and cable tasks.
 - Delivered Isaac GROOT N1.7 data conversion, model training, and policy inference, separately validating virtual-data training followed by real-world inference and real-data training/inference workflows.
 - Built a two-part training-data collection flow: an automated workflow ran scripts to control the robot arm and task sequence, while a separate recording system captured actions, images, robot state/actions, and timestamps. They communicated through ROS 2, WebSocket, and related interfaces to fully automate multiple imitation-learning / GROOT / VLA training episodes, achieving 3-4x manual throughput and reducing manual staffing needs by two operators.
 - Owned robotics software from requirements and architecture through Docker deployment, testing, on-site integration, and maintenance; used YOLO OBB with OpenCV for HDMI insertion and AprilTag with OpenCV for HDMI / power-cable unplugging, while cable-insertion success improved from 70% to 90% in a 10-trial test.
 - Integrated wrist-mounted six-axis force/torque sensing and force-guided contact search into the final HDMI insertion stage; when residual visual alignment error remained, used low-speed contact and bounded spiral micro-search to recover insertion. Validated on the physical robot; incremental success-rate impact is not yet quantified.
- ISCOM | Software Engineer InternFeb 2022 - Jun 2022 | Kaohsiung, Taiwan
- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Robot Control & Digital Twin	ROS 2, NVIDIA Isaac Sim / Lab, MoveIt 2, cuMotion, virtual / physical robot workflows
VLA & Robot Learning	Isaac GROOT N1.7, VLA, Policy Inference, Imitation Learning, Robot State / Action
Perception & Visual Inspection	RealSense, YOLO OBB + OpenCV (HDMI insertion), AprilTag + OpenCV (HDMI / power-cable unplugging), Hand-Eye Calibration, Visual Servoing, wrist-mounted six-axis force/torque sensing, force-guided contact search, PatchCore anomaly detection prototype with PyQt
Software & System Integration	Python, C++, frontend/backend platforms with FastAPI, React / TypeScript, Next.js, WebSocket, Docker, Linux, Arduino, ESP32, MQTT

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place